Dealer-Gamma Feedback and Local Volatility Cycles: A Fixed-Equilibrium Bifurcation Model

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Abstract

We construct a local physical-measure feedback model whose equilibrium remains fixed as coupling varies and whose variance drift points inward at zero. For a centered Gaussian density of signed dealer positions, the local Hopf equation is quadratic in coupling. An illustrative, non-calibrated configuration gives $\kappa^\star = 31.4933$ yr^{-1}, $\omega^\star = 47.1186$ rad yr^{-1}, and $\ell_1 = -6.2888$; the actual nonlinear book produces a nearby attracting cycle with positive variance. Public open interest does not identify dealer sign, so the pre-extraction WRDS protocol is an unsigned registered-horizon proxy study, not a structural confirmation. No registered market dataset is analysed.

1 Model and measurement boundary

Option hedging can create state-dependent underlying demand [Frey and Stremme, 1997, Sircar and Papanicolaou, 1998]. Demand-based option pricing [Gârleanu et al., 2009] does not itself derive a spot drift, so our feedback is a reduced-form physical-measure hypothesis. Let $X_t = \log(S_t/F_t)$ for a predictable reference and specify

$$
\begin{aligned}
dX_t &= [-\delta X_t - \tfrac{1}{2}(v_t - \theta_v) + \kappa g(X_t, v_t, \chi_t)]dt + \sqrt{v_t}\, dW_t^S, && (1)\\
dv_t &= [\kappa_v(\theta_v - v_t) + \gamma v_t \chi_t]dt + \xi\sqrt{v_t}\, dW_t^v, && (2)\\
d\chi_t &= \alpha(\beta X_t - \chi_t)dt. && (3)
\end{aligned}
$$

At $v = 0$, drift is $\kappa_v\theta_v > 0$ and diffusion is zero. Thus the added feedback is boundary-compatible, although we do not prove global non-explosion.

A Gaussian mass in fixed log moneyness integrates against Black–Scholes gamma as

$$
\begin{aligned}
\mathcal{G}(X, v) &= \{2\pi\tau^2(v)\}^{-1/2} e^{-q_d T - X} \exp[-(X - m(v))^2/(2\tau^2(v))],\\
\tau^2(v) &= \sigma_q^2 + vT, \qquad m(v) = \mu_q - (r - q_d + v/2)T. && (4)
\end{aligned}
$$

We center and normalize it:

$$
g(X, v, \chi) = s\frac{\mathcal{G}(X, v) - \mathcal{G}(0, \theta_v)}{\mathcal{G}(0, \theta_v)}, \qquad s \in \{-1, +1\}. \tag{5}
$$

Then $(0, \theta_v, 0)$ is an equilibrium for every κ. The orientation s describes latent signed dealer positions. Public OI counts contracts but does not identify participant or side [Options Industry Council, 2025]; proprietary participant-tagged data are used when actual market-maker positions are needed [Vasquez et al., 2025].

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2 Local bifurcation

At the fixed equilibrium, put $G_X = g_X$, $G_v = g_v$, $G_\chi = g_\chi$. The Jacobian is

$$J = \begin{pmatrix} a & b & d \\ 0 & -\kappa_v & c \\ m & 0 & -\alpha \end{pmatrix}, \quad \begin{matrix} a = -\delta + \kappa G_X, & b = -1/2 + \kappa G_v, & \\ d = \kappa G_\chi, & c = \gamma\theta_v, & m = \alpha\beta. \end{matrix} \tag{6}$$

For $\det(\lambda I - J) = \lambda^3 + c_2\lambda^2 + c_1\lambda + c_0$,

$$c_2 = \kappa_v + \alpha - a, \quad c_1 = \kappa_v\alpha - a(\kappa_v + \alpha) - dm,$$
$$c_0 = -a\kappa_v\alpha - bcm - dm\kappa_v.$$

Theorem 1. *If the drift is C^3 and, at $\kappa^\star > 0$, $c_2, c_1, c_0 > 0$, $H := c_1c_2 - c_0 = 0$, $H' \neq 0$, and $\ell_1 \neq 0$, then the fixed equilibrium undergoes a non-degenerate local Hopf bifurcation. Its eigenvalues are $\pm i\sqrt{c_1}$ and $-c_2$, with*

$$\left.\frac{d\,\mathrm{Re}\,\lambda_+}{d\kappa}\right|_{\kappa^\star} = -\frac{H'(\kappa^\star)}{2(c_1 + c_2^2)}.$$

In the Kuznetsov [2004] convention, $\ell_1 < 0$ gives stable small cycles on the side where the equilibrium pair is unstable.

For (5), $G_\chi = 0$. With $A = \alpha + \kappa_v$, $M = \alpha\kappa_v$, and $L = (\gamma\theta_v)(\alpha\beta)$,

$$H(\kappa) = G_X^2 A\kappa^2 + \{G_v L - G_X A(A + 2\delta)\}\kappa + A(M + \delta A + \delta^2) - L/2. \tag{7}$$

Every root still requires the three positivity conditions and transversality. Negative discriminant means only no local Hopf root. Analytic derivatives of (4) through order three enter the standard first-Lyapunov formula; the variance tensor includes $B_{v,v\chi} = B_{v,\chi v} = \gamma$.

3 Actual-nonlinearity validation

For the illustrative, non-fitted configuration $(\delta, \kappa_v, \theta_v, \alpha, \beta, \gamma) = (0.5, 40, 0.0625, 50, 2, 500)$ and $(\mu_q, \sigma_q, T, s) = (0.06, 0.20, 0.25, +1)$, $G_X = -0.0617978$ and $G_v = -2.0198878$. Enumerating both quadratic roots gives

$$\kappa^\star = 31.4932976, \qquad \omega^\star = 47.1185670, \qquad \ell_1 = -6.2888041,$$

with crossing speed 0.2686294. The period is 0.13335 years. The remote second valid root is $\kappa^{\star\star} = 16860.8961$, where the pair crosses back into the stable half-plane; threshold plots use the first root.

Direct integration of (1)–(3) with the actual Gaussian book gives $X \in [-0.02814, 0.02358]$ and $v \in [0.04453, 0.08187]$ after transient discard. The displayed cycle therefore stays in the local, positive-variance region. This validates an internally consistent possibility, not SPX calibration or global stability. Gross-normalized signed mixtures in the full paper show that supercriticality is book-dependent; dispersed or offsetting books can instead be subcritical.

4 Pre-extraction protocol and limitations

Before expected September 2026 WRDS access, Amendments A13–A15 fix the registered historical horizon from 2017-01-03 through 2024-10-29. Four observable OI-gamma summaries—log unsigned mass, call–put composition, mean log moneyness, and log dispersion—separately predict $\log(r_{t+1}^2 + 10^{-8})$ with leads and HAR-style lags constructed on the complete CRSP market calendar, official Cboe VIX, outcome-session weekday indicators, and one regressor-session monthly-expiration control. Contracts are unadjusted standard-settlement SPX/SPXW with 1–365 calendar-day maturities, fixed quote/IV/moneyness filters, and parity forwards split by AM/PM settlement. HAC lag 5 and a 10-day moving-block bootstrap are separately BH-adjusted over four tests. Robust association requires rejection by both adjusted families and a 95% bootstrap interval excluding zero; one-family rejection is method-sensitive. Convention-signed GEX and stress interactions are secondary. This is a retrospective pre-analysis plan over historically public outcomes, not a blinded prospective experiment.

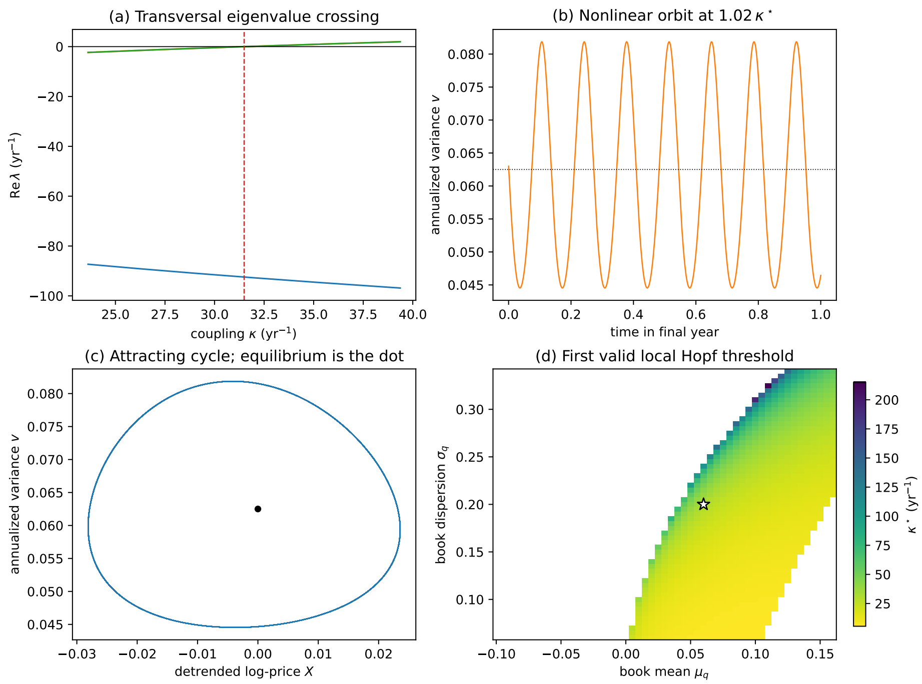

Figure 1: Eigenvalue crossing, actual nonlinear orbit at $1.02\kappa^{\star}$, and local-Hopf map. Blank cells do not imply stability.

The theorem is local and deterministic. Brownian paths are not periodic orbits. The model has no proved global attractor, stationary tail result, stochastic threshold shift, Hawkes equivalence, or causal identification. For an affine additive linearization, the tangent cocycle is e^{Jt} and is noise-independent; the earlier nonzero shift claim was withdrawn. The full square-root model has state-dependent diffusion and is not analysed as a stochastic bifurcation here. Any WRDS association concerns public book summaries. Confirming signed dealer feedback requires participant-tagged positions and an identification strategy.

References

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